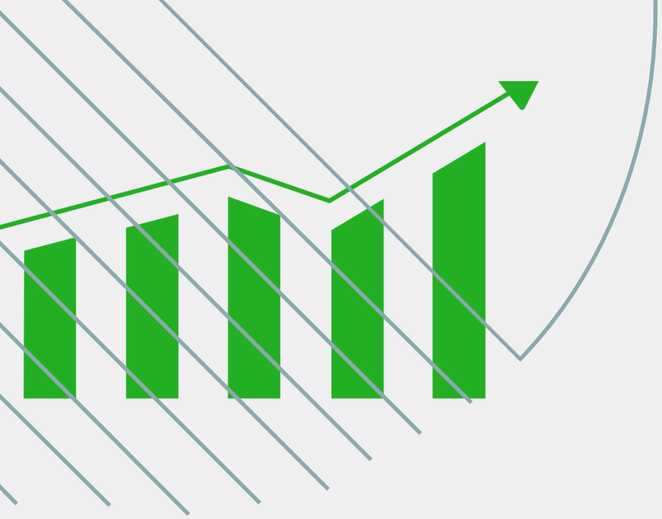
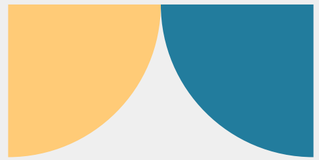


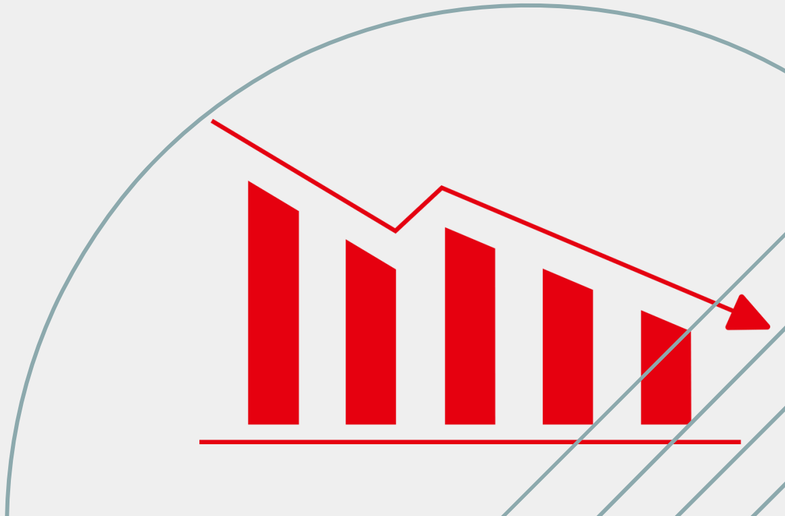


STOCK MARKET PREDICTION



Predicting Stock Prices using
Machine Learning







THE STOCK MARKET

DATASET OVERVIEW

We used historical stock data for NFLX, which includes the following features: 'Open,' 'High,' 'Low,' 'Close,' and 'Volume.'

The dataset covers daily stock prices, providing a comprehensive view of the market's fluctuations over time.

	A	B	C	D	E	F
1	Date	Open	High	Low	Close	Volume
2	05/31/2024	644.01	648.58	628.3	641.62	4,071,023
3	05/30/2024	653.52	656.49	644.73	647.66	2,393,354
4	05/29/2024	644.13	664.25	643.89	654.62	3,695,935
5	05/28/2024	647.05	649.75	643.03	649	2,609,451

PROBLEM STATEMENT & APPROACH

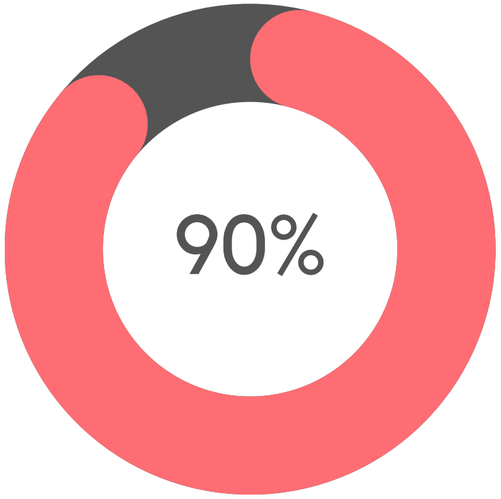
- The goal of our project is to accurately predict the 'Open' price of NFLX (Netflix) stock using historical stock data.
- The 'Open' price is the price at which a stock first trades upon the opening of the stock market on a given day.
- Accurate prediction of this value is crucial for traders and investors, as it can influence trading strategies and investment decisions.

MODEL SELECTION

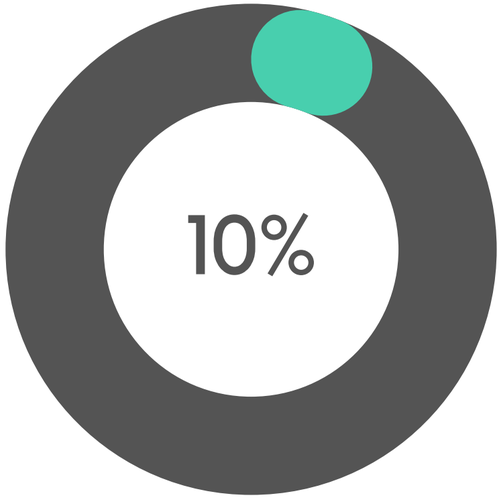
We experimented with various machine learning models to predict the stock prices, including:

01 - LINEAR REGRESSION	A simple yet powerful model for continuous prediction.
02 - RANDOM FOREST	An ensemble of decision trees that improves accuracy.
03 - DECISION TREE	A model that splits data into branches for prediction.
04 - GRADIENT BOOSTING	A model that builds trees sequentially to minimize errors.
05 - SUPPORT VECTOR REGRESSION (SVR)	A model that tries to fit the best line within a threshold.

TRAINING THE MODELS



TRAINING



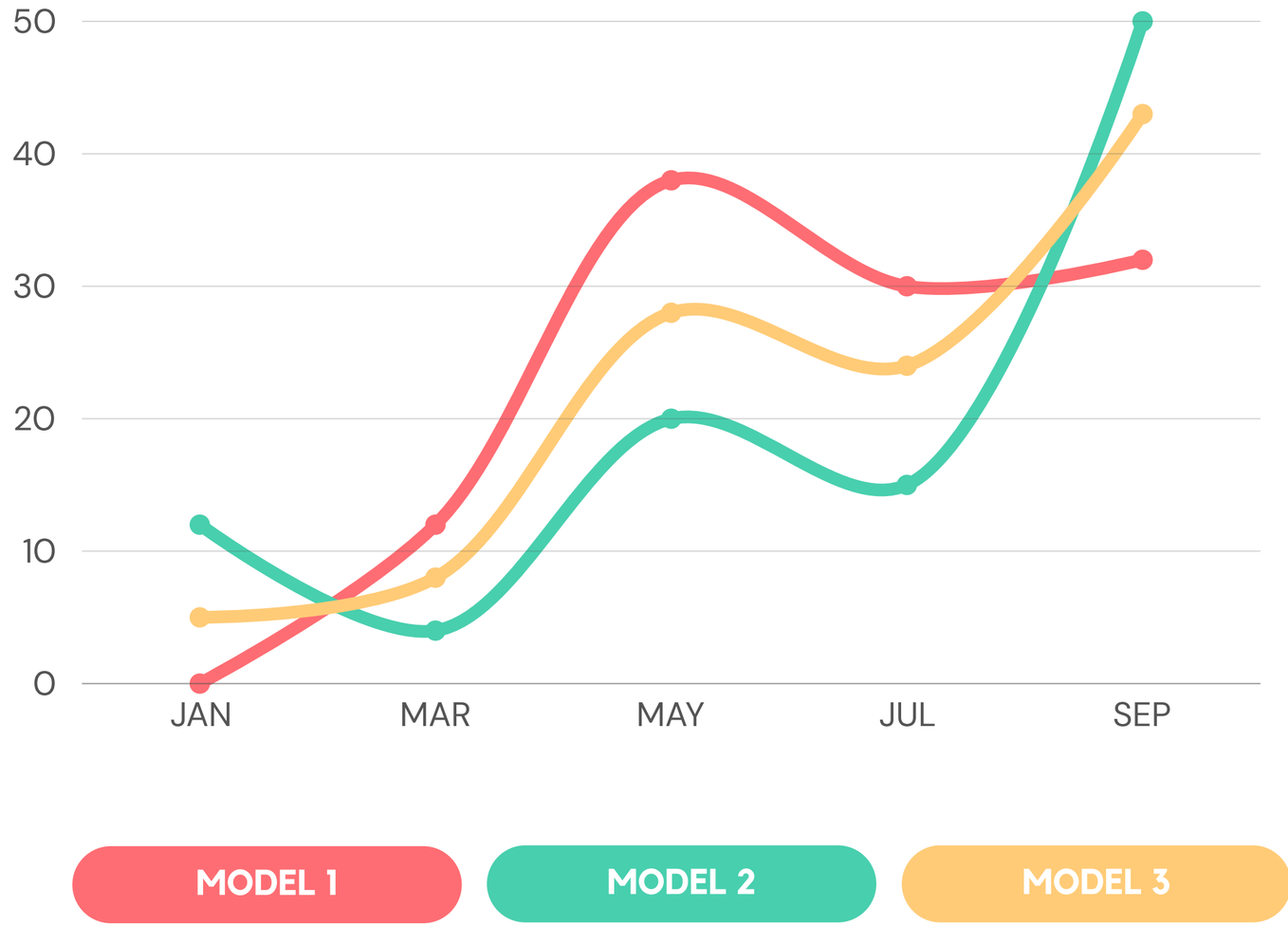
TESTING

The dataset was split into training and testing sets with a 90-10 split ratio.

MODEL EVALUATION

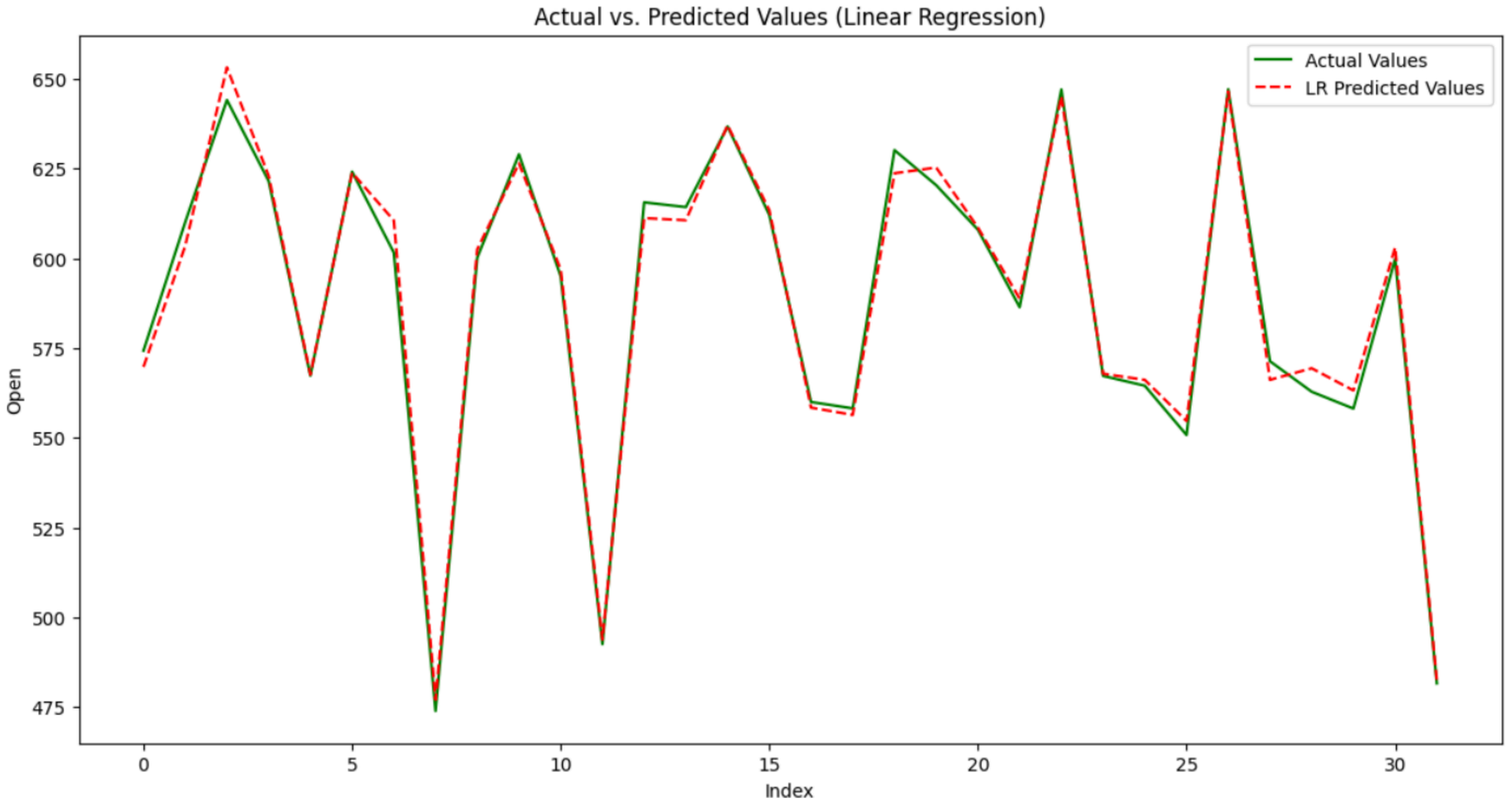
The models were evaluated based on their prediction accuracy using the test data

We used the model score (R^2) as the primary evaluation metric



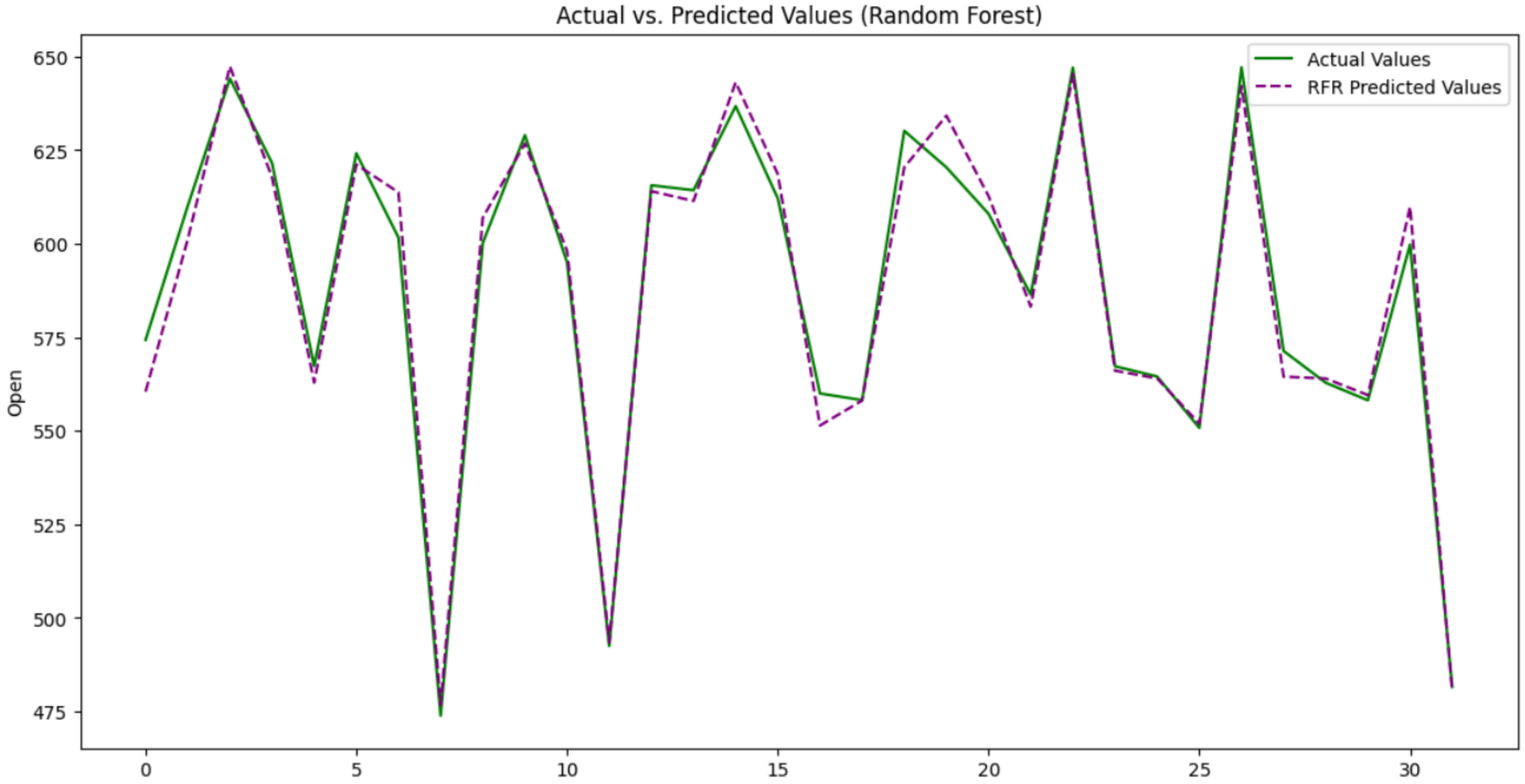
RESULTS

01 - LINEAR REGRESSION



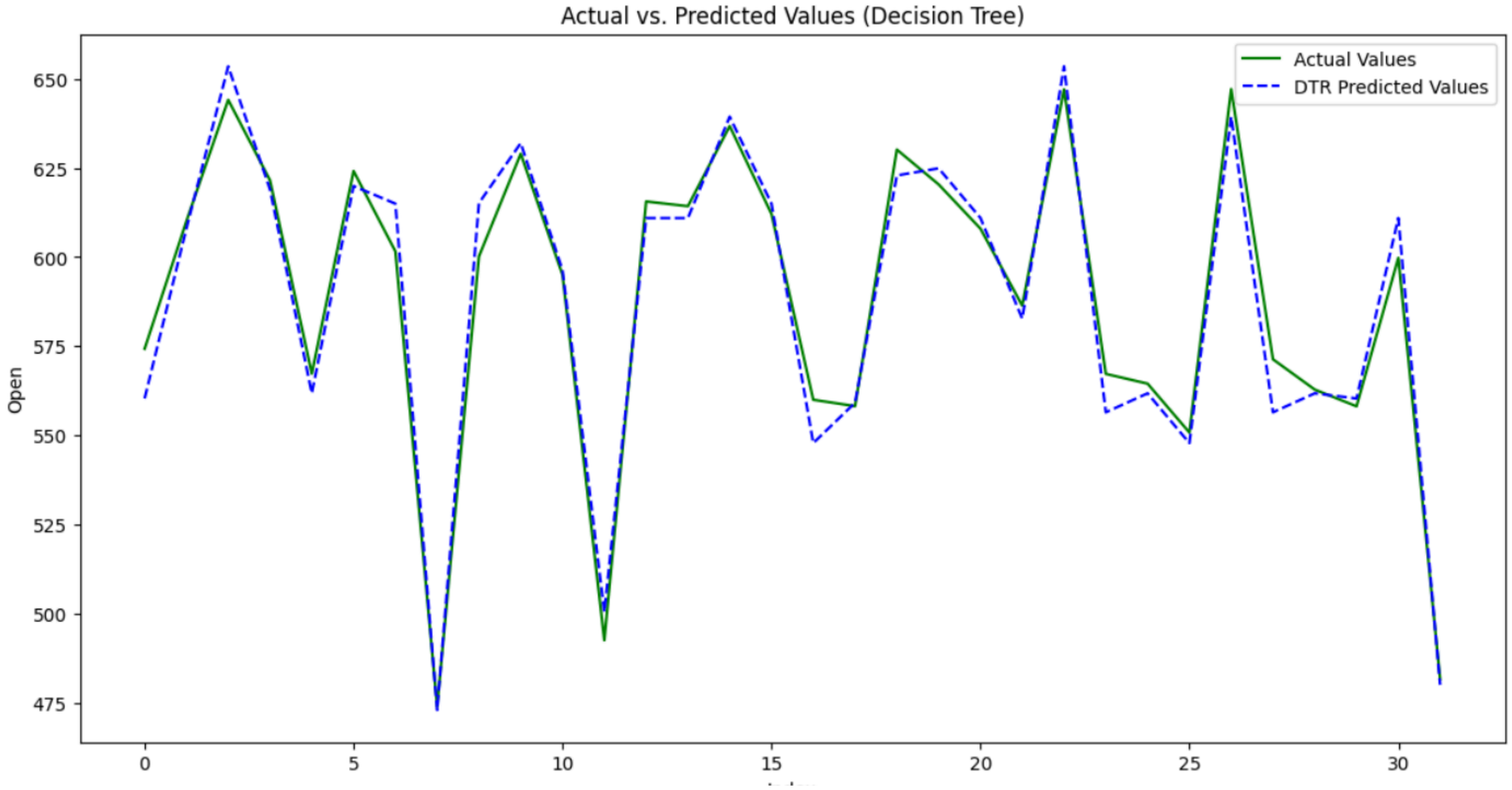
RESULTS

02 - RANDOM FOREST



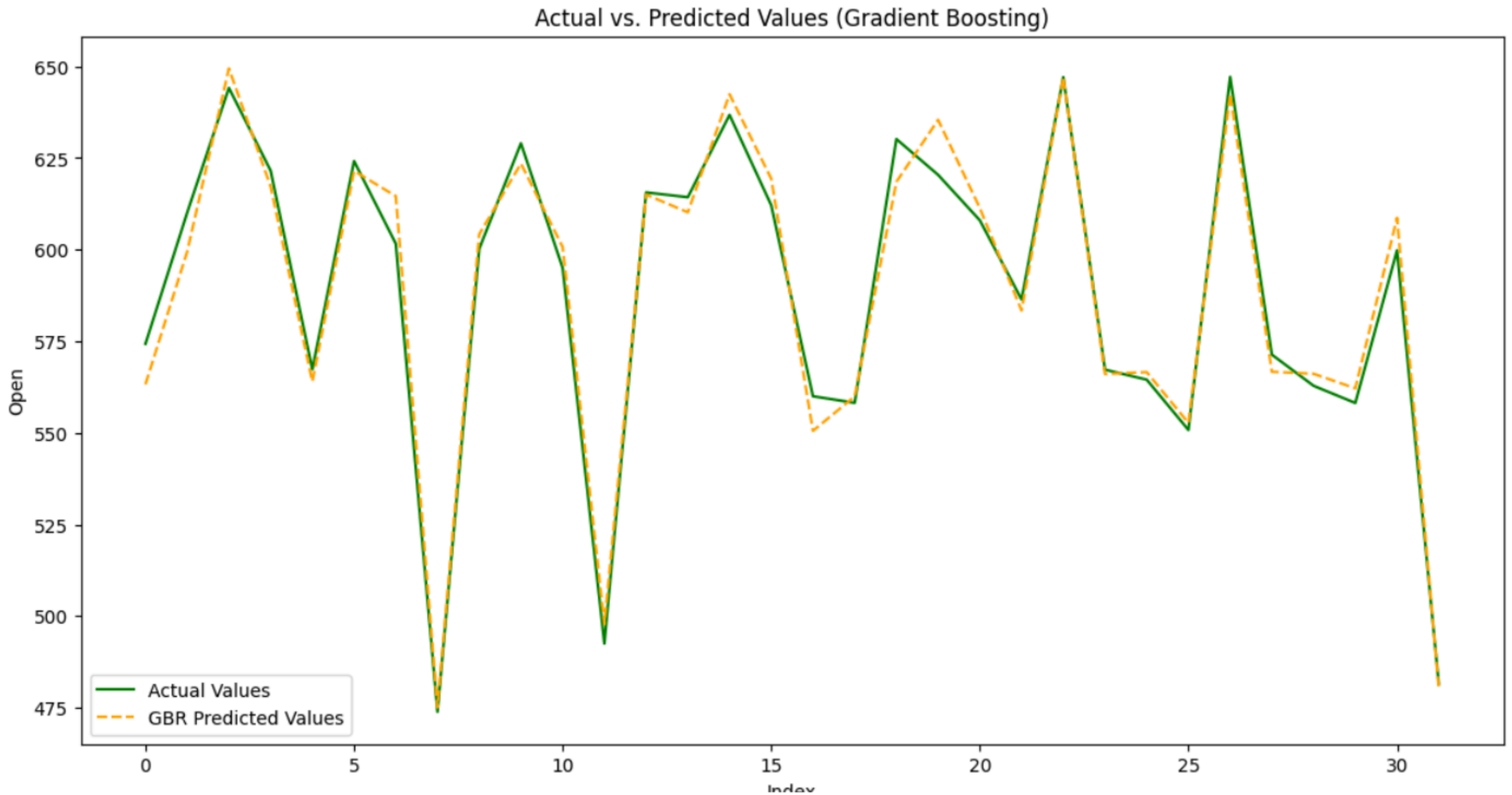
RESULTS

03 - DECISION TREE



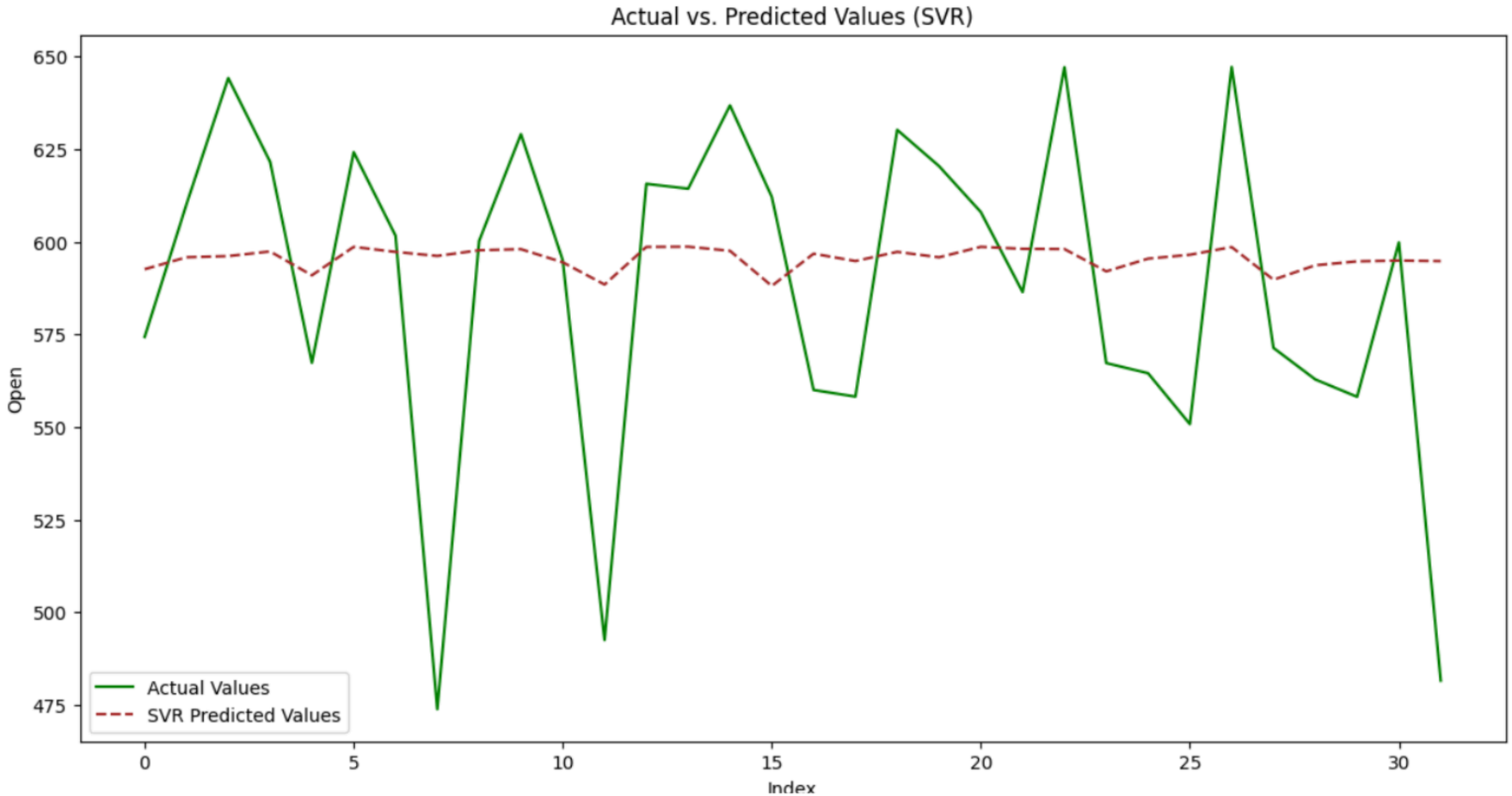
RESULTS

04 - GRADIENT BOOSTING

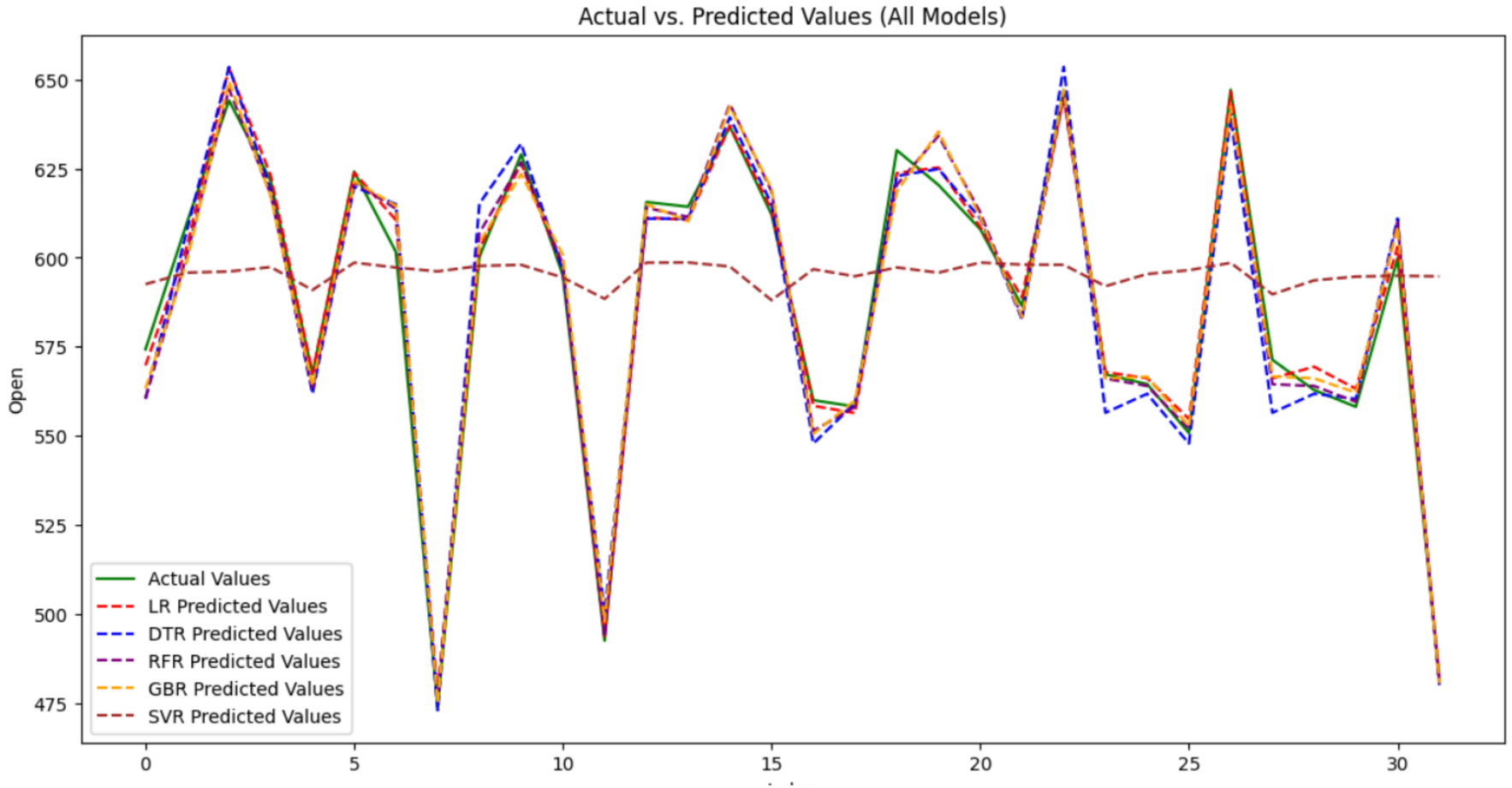


RESULTS

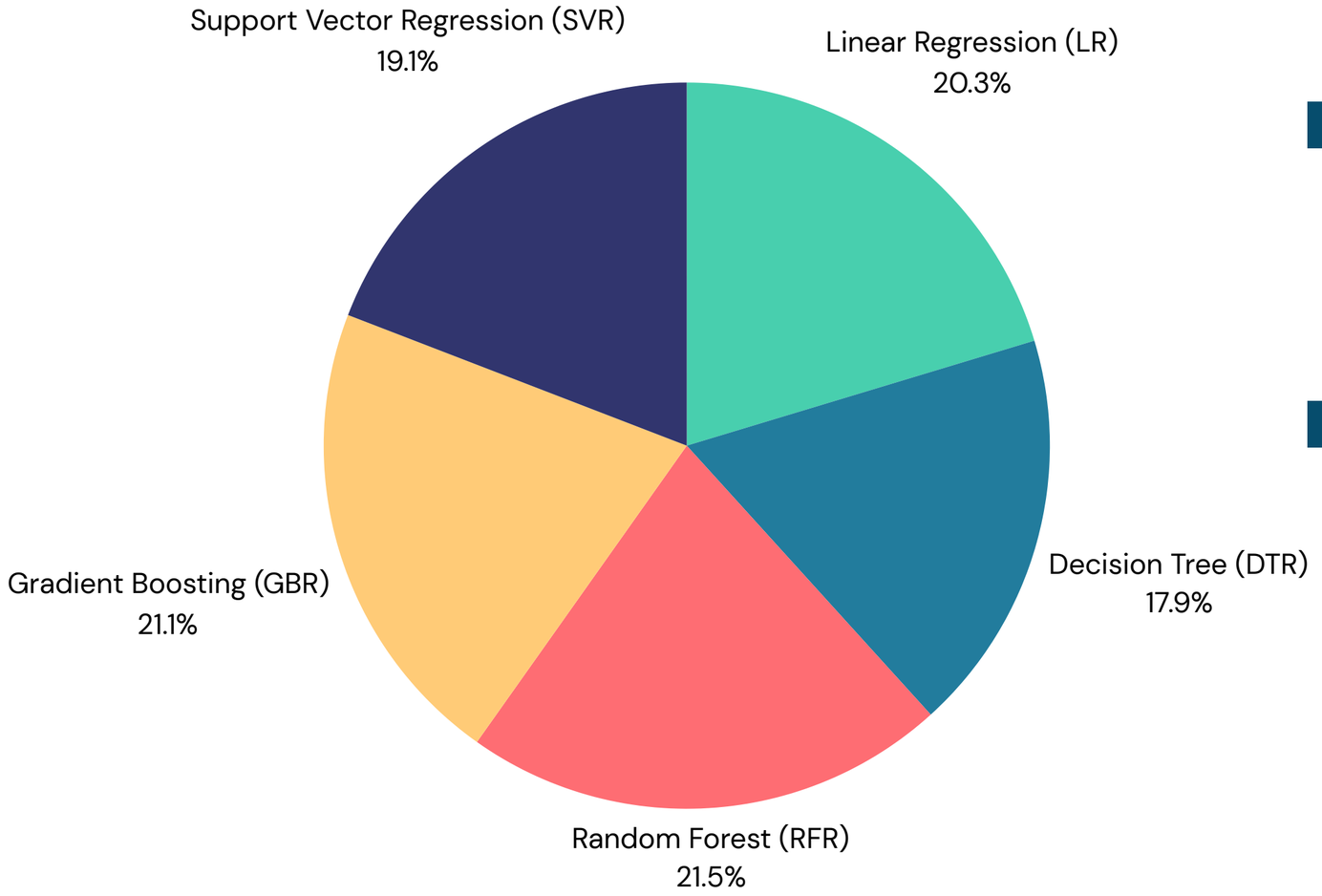
05 - SUPPORT VECTOR REGRESSION (SVR)



RESULTS (All Models)



OVERALL RESULT



- Linear Regression and Random Forest performed well, closely matching the actual prices
- Gradient Boosting also showed promising results, while Decision Tree and SVR had some variance.

SUMMARY

Random Forest provided the best balance of accuracy and stability, making it the most reliable model for our dataset

THANK YOU

"STOP DREAMING
AND START
DOING"